

Who Gets the Blame?

A Structural Theory of Scapegoating in Social Networks

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Scapegoating



The act of shifting blame onto innocent individuals or groups

Examples



Witch Hunts



Ethnic Purges



Immigrant Blame



Corporate Scandals

Research Gap

- **Fact:** Scapegoating often targets **marginalized** individuals
- **Existing research:** **Identity-based** explanation (who people are)
 - Gender
 - Ethnicity
 - Religion
 - Partisanship
 - Nationality
 - In-group vs. out-group
- **This study:** **Network-based** explanation (where individuals are positioned within social structures)

Research Gap

- **Relevant network perspective:**
Decentralized outcomes of peer interactions
 - Negative and conflictual ties
 - Bullying and victimization networks
 - Social control and punishment in networks
- **This study:**
A **strategic, leader-driven** mechanism using a game-theoretic approach



*"The best way to bring folks together is to give them a really good enemy."
— The Wizard, in Wicked*

The Model

Model setup:

- A community:
 - A leader L
 - A total of n agents embedded in an undirected graph $G(V, E)$
- Two possible states for a crisis:
 - $\theta_i = 0$: Agent i is innocent $\implies$ For all $i \in V$, $\theta_i = 0$
 - $\theta_i = 1$: Agent i is guilty

Phase 1: Scapegoat Selection

- **Leader's strategies:**

$$s_L : G \rightarrow \{\emptyset\} \cup V$$

- $s_L(G) = \emptyset$: Self-blame (game ends)
- $s_L(G) = k \in V$: Scapegoat some agent k (game proceeds)

- **Leader's utilities:**

$$v_L(s_L) = \begin{cases} -C & \text{if } s_L(G) = \emptyset \\ -R^{(k)} & \text{if } s_L(G) = k \text{ for some } k \in V \end{cases}$$

- C : Fixed cost of self-blaming
- $R^{(k)} = \sum_{i=1}^n (1 - x_i^{*(k)})$: Reputational cost of scapegoating (determined in Phase 2)

Phase 2: Opinion Dynamics

- **Agents' beliefs:**

- $\pi_i^{(k)}$: Agent i 's prior belief of agent k 's guilt ($\theta_k = 1$)
- $p \in [0, 1]$: Probability that the leader prosecutes an innocent agent $\implies$ public trust
- $b_i^{(k)} = \frac{\pi_i^{(k)}}{\pi_i^{(k)} + p(1 - \pi_i^{(k)})}$: Agent i 's internal posterior belief about agent k 's guilt

- **Agents' strategies:**

- $x_i^{(k)} \in [0, 1]$: Agent i 's public opinion about agent k 's guilt through social learning

- **Agents' utilities:**

$$u_i(x_i^{(k)}, x_{-i}^{(k)}) = \underbrace{-(x_i^{(k)} - b_i^{(k)})^2}_{\text{Penalty 1: Self fidelity}} - \underbrace{\sum_{j=1}^n A_{ij}(x_i^{(k)} - x_j^{(k)})^2}_{\text{Penalty 2: Peer conformity}}$$

- A game-theoretic version of opinion dynamics (Ghaderi & Srikant, 2014)

Equilibrium Analysis

- Phase 2:

$$x^{*(k)} = (I + L)^{-1} b^{(k)}$$

- Phase 1:

$$s_L^*(G) = \begin{cases} \emptyset & \text{if } R^{(k)} > C \text{ for any } k \in V \\ \arg \min_{k \in V} R^{(k)} & \text{if } R^{(k)} \leq C \text{ for some } k \in V \end{cases}$$

Lemma (Opinion Conservation)

For all $k \in V$, $\sum_{i=1}^n x_i^{*(k)} = \sum_{i=1}^n b_i^{(k)}$.

- Laplacian matrix L : Row stochastic
- **Intuition: The overall “mass” of opinions remains unchanged**
- $R^{(k)} = \sum_{i=1}^n (1 - x_i^{*(k)}) = \sum_{i=1}^n (1 - b_i^{(k)}) = \sum_{i=1}^n [1 - \frac{\pi_i^{(k)}}{\pi_i^{(k)} + p(1 - \pi_i^{(k)})}]$
 $\implies$ Determined by the social structure of $\pi_i^{(k)}$ (**network effects!**)

Discrete Network Effects

Definition 1 (Discrete Network Effects)

$$\pi_i^{(k)} = \begin{cases} 0 & \text{if } i \in N(k), \\ \frac{1}{2} & \text{if } i \notin N(k). \end{cases}$$

$$\Rightarrow R^{(k)} = \sum_{i \notin N(k)} \left(1 - \frac{1}{1+p}\right) + \sum_{i \in N(k)} (1 - 0) = \frac{np + \text{deg}(k)}{1+p}$$

Theorem 1

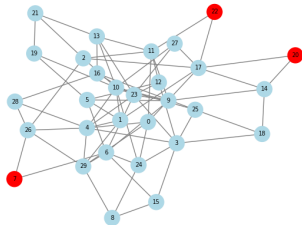
Given discrete network effects, if there exists some $k \in V$ such that $R^{(k)} \leq C$, then $s_L^*(G) = \arg \min_{k \in V} \text{deg}(k)$.

- The leader scapegoats the agent with the lowest **degree centrality**
 $\Rightarrow$ **Local structural vulnerability**

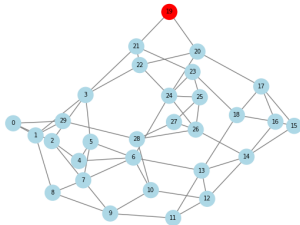
Discrete Network Effects

Discrete Network Effects — Scapegoat Selection ($p=0.3$)

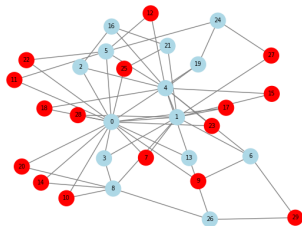
ER ($n=30$)
Selected: [7, 20, 22]



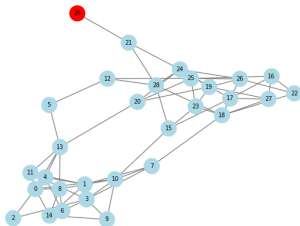
WS ($n=30$)
Selected: [19]



BA ($n=30$)
Selected: [7, 9, 10, 11, 12, 14, 15, 17, 18, 20, 22, 23, 25, 27, 28, 29]



SBM ($n=30$)
Selected: [29]



Decay Network Effects

Definition 2 (Decay Network Effects)

Let l_{ik} denote the geodesic (shortest-path distance) between i and k . Define

$$\pi_i^{(k)} = \frac{1}{2} - \left(\frac{1}{2}\right)^{l_{ik}} \begin{cases} = 0 & \text{if } l_{ik} = 1, \\ \rightarrow \frac{1}{2} & \text{if } l_{ik} \rightarrow \infty. \end{cases}$$

Theorem 2

Given decay network effects, if there exists some $k \in V$ such that $R^{(k)} \leq C$, then $s_L^*(G) = \arg \min_{k \in V} D_k$, where D_k is a metric determined by Algorithm 1.

⇒ Global structural vulnerability

Algorithm 1 Scapegoat Selection Algorithm with Decay Network Effects

Require: Graph $G(V, E)$, parameter p

Ensure: Selected nodes with the lowest D_k

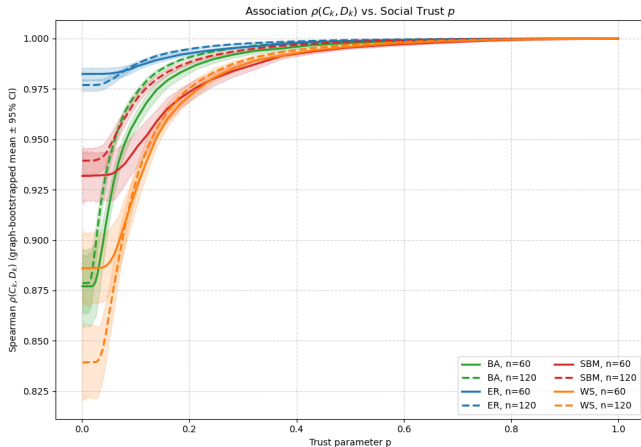
- 1: **for** each node $k \in V$ **do**
 - 2: Initialize vector $\pi_k \in \mathbb{R}^{|V|}$
 - 3: **for** each node $i \in V$, $i \neq k$ **do**
 - 4: Calculate the shortest path length l_{ki} between k and i using **Breadth-First Search (BFS)** for unweighted G and **Dijkstra's or Bellman-Ford Algorithm** for weighted G
 - 5: $\pi_i^{(k)} \leftarrow 0.5 - 0.5^{l_{ki}}$
 - 6: **end for**
 - 7: $\pi_k^{(k)} \leftarrow 0$
 - 8: Generate vector b_k by $b_i^{(k)} = \frac{\pi_i^{(k)}}{\pi_i^{(k)} + p(1 - \pi_i^{(k)})}$ where $\forall i \in V$.
 - 9: Calculate $D_k = \sum_{i \in V} (1 - b_i^{(k)})$
 - 10: **end for**
 - 11: Select the nodes with the lowest D_k values
 - 12: **Output** the selected nodes
-

Is D_k equivalent to **decay centrality**?

Decay Network Effects

Proposition

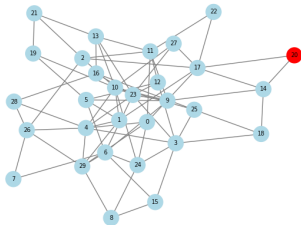
The metric D_k is a component-wise convex transformation of decay centrality $C_k = \sum_{i=1}^n \alpha^{l_{ik}}$ with $\alpha = \frac{1}{2}$. Specifically, when $p = 1$, $D_k \sim C_k$.



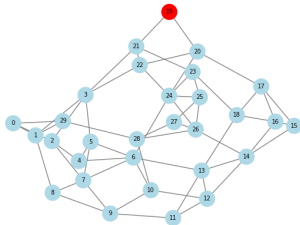
Decay Network Effects

Decay Network Effects — Scapegoat Selection ($p=0.3$)

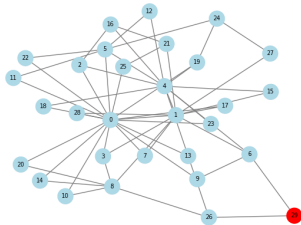
ER ($n=30$)
Selected: [20]



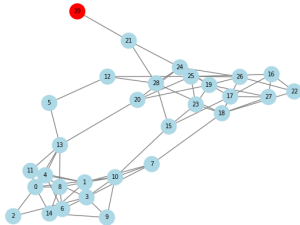
WS ($n=30$)
Selected: [19]



BA ($n=30$)
Selected: [29]



SBM ($n=30$)
Selected: [29]



Extension 1: From Equilibrium to Complexity

- **So far:** Deterministic network effects
 - Discrete: Local structural vulnerability
 - Decay: Global structural vulnerability
- **Next:** Does structural vulnerability persist in stochastic and heterogeneous settings?

Relaxation 1 (Stochastic Network Effects)

$$\pi_i^{(k)} = \sigma(\beta_0 + \beta_1 l_{ik} + \varepsilon_i), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_\pi^2)$$

- $\sigma(\cdot)$: Logistic function $\sigma(z) = \frac{1}{1+e^{-z}}$
- β_0 : Baseline skepticism
- $\beta_1 > 0$: Slope of distance-based skepticism
- σ_π : Noise level (idiosyncratic randomness)

Extension 1: From Equilibrium to Complexity

Relaxation 2 (Heterogeneous Public Trust)

$$p_i = \sigma(\mu_p + \eta_i), \quad \eta_i \sim \mathcal{N}(0, \sigma_p^2)$$

- $\sigma(\cdot)$: Logistic function $\sigma(z) = \frac{1}{1+e^{-z}}$
- μ_p : Average trust level
- σ_p : Dispersion of trust heterogeneity

Relaxation 3 (Heterogeneous Conformity/Stubbornness)

$$u_i(x_i, x_{-i}) = -(1 - \alpha_i)(x_i - b_i)^2 - \alpha_i \sum_{j=1}^n A_{ij}(x_i - x_j)^2, \quad \alpha_i \sim U(0, 1)$$

Relaxation 4 (Stochastic Decision)

$$P(k) \propto \exp[-\lambda(R^{(k)} - \min_j R^{(j)})]$$

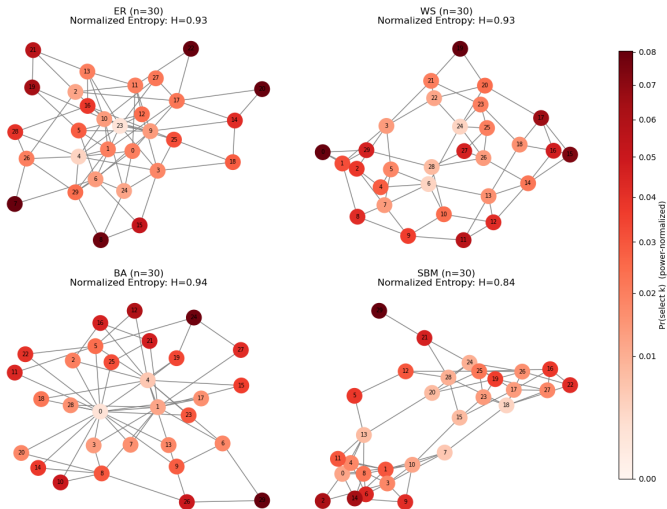
- $\lambda > 0$: Parameter for bounded rationality; $\lambda \rightarrow \infty$: Deterministic decision

Extension 1: From Equilibrium to Complexity

Stochastic Network Effects — Scapegoat Selection Frequency Heatmaps

Simulation settings:

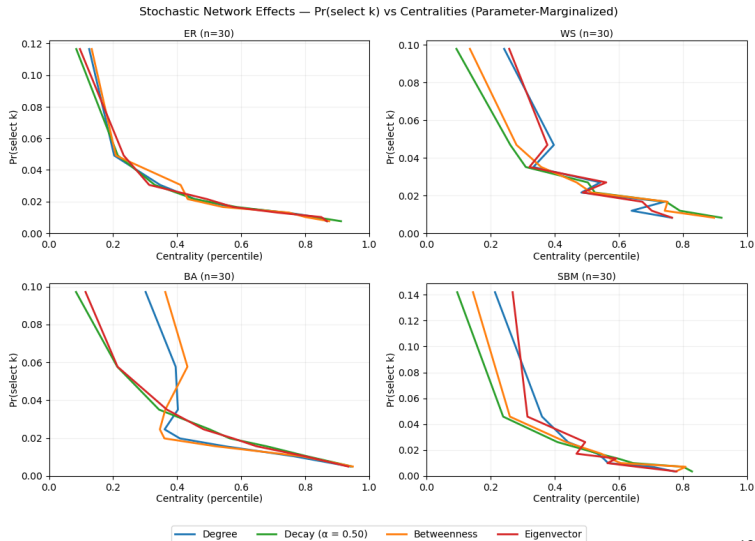
- Trials per parameter:
500 Monte Carlo runs
- $\beta_0 = -2.0$
- $\beta_1 = 0.3$
- $\sigma_\pi = 1.0$
- $\mu_p = 0.0$
- $\sigma_p = 1.5$
- $\lambda = 15.0$



Extension 1: From Equilibrium to Complexity

Simulation settings:

- Graphs per model: 5
- Trials per parameter: 500 Monte Carlo runs
- $\beta_0 = -2.0$
- $\beta_1 \in \{0.2, 0.3, 0.6\}$
- $\sigma_\pi \in \{0.5, 1.0, 1.5\}$
- $\mu_p = 0.0$
- $\sigma_p = 1.5$
- $\lambda = 15.0$

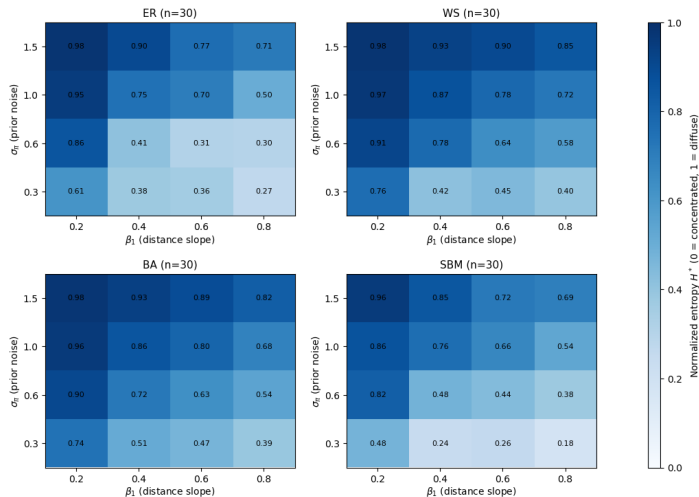


Extension 1: From Equilibrium to Complexity

Simulation settings:

- Graphs per model: 10
- Trials per parameter: 1000 Monte Carlo runs
- $\beta_0 = -2.0$
- $\beta_1 \in \{0.2, 0.4, 0.6, 0.8\}$
- $\sigma_\pi \in \{0.3, 0.6, 1.0, 1.5\}$
- $\mu_p = 0.0$
- $\sigma_p = 1.5$
- $\lambda = 15.0$
- $H^* = \frac{-\sum_k P(k) \log P(k)}{\log n}$

Stochastic Network Effects — System-Level Normalized Entropy H^* over (β_1, σ_π)



Extension 2: Identity-Structure Interaction

Relaxation 5 (Heterogeneous Identity)

$$\text{logit}(\tilde{\pi}_i^{(k)}) = \text{logit}(\pi_i^{(k)}) + \gamma \mathbf{1}\{g_i = \text{in}, g_k = \text{out}\}$$

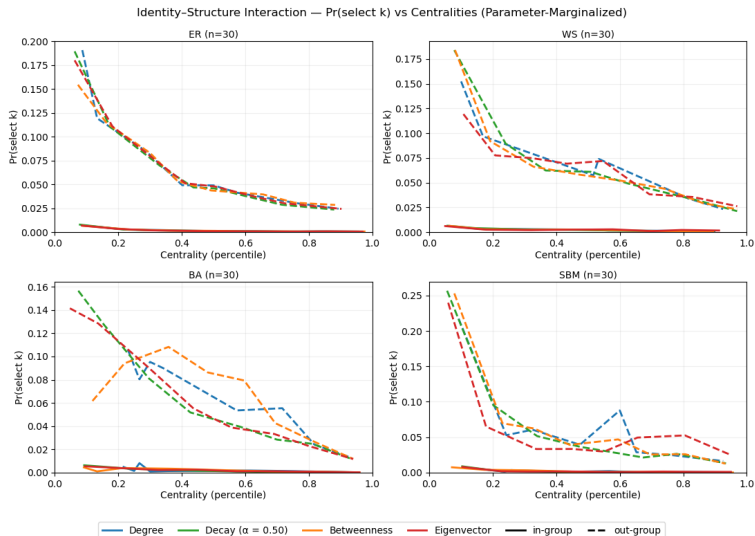
$$\Leftrightarrow \tilde{\pi}_i^{(k)} = \begin{cases} \frac{e^{\gamma \pi_i^{(k)}}}{(1 - \pi_i^{(k)}) + e^{\gamma \pi_i^{(k)}}}, & \text{if } g_i = \text{in}, g_k = \text{out}, \\ \pi_i^{(k)}, & \text{otherwise.} \end{cases}$$

- $\text{logit}(p) = \log \frac{p}{1-p}$
- $\tilde{\pi}_i^{(k)}$: Biased prior belief with identity-based distortion
- γ : Strength of social identity bias
- $g_i \in \{\text{in}, \text{out}\}$: Group label of agent i

Extension 2: Identity-Structure Interaction

Simulation settings:

- Graphs per model: 5
- Trials per parameter: 500 Monte Carlo runs
- $\beta_0 = -2.0$
- $\beta_1 \in \{0.2, 0.3, 0.6\}$
- $\sigma_\pi \in \{0.5, 1.0, 1.5\}$
- $\mu_p = 0.0$
- $\sigma_p = 1.5$
- $\lambda = 15.0$
- $\gamma = 1.0$

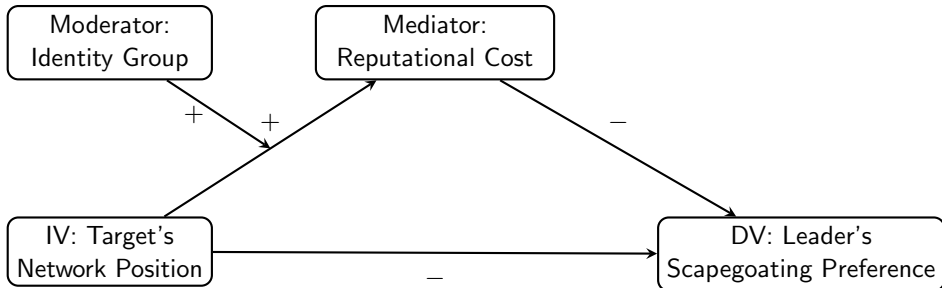


Ongoing Experiment

- **Experimental design:**

- Simulate a crisis scenario in an organizational context
- Participants assume the role of a leader
- Provide diagrams of team communication networks, varying the target's position

- **Conceptual model:**



Conclusion

Insights:

- Social psychology
- Political science
- Organizational behavior
- Network economics
- Complexity science

Significance:

- Distinguish:
 - Structural inequality
 - Social identity bias
- Intervention strategies:
 - Informational (e.g., transparency)
 - Structural (e.g., social ties, cross-group connectivity)

Future work:

- Endogenous identity-structure interactions
- Dynamic leadership adaptation and network co-evolution

The End